Frontiers of Science

**Homework Questions for Climate and Us Week 1**
**Due the week of November 17**
**ANSWER KEY**

## 1. Earth's Energy Balance

Any given planet's climate is determined by its energy balance. Earth's energy balance is such that it creates a livable environment for humans and other life, unlike those of nearby Venus or Mars. Climate forcings are factors that drive changes in climate and alter the Earth's energy balance. **Use Figure 1 (below) to answer the following questions.**

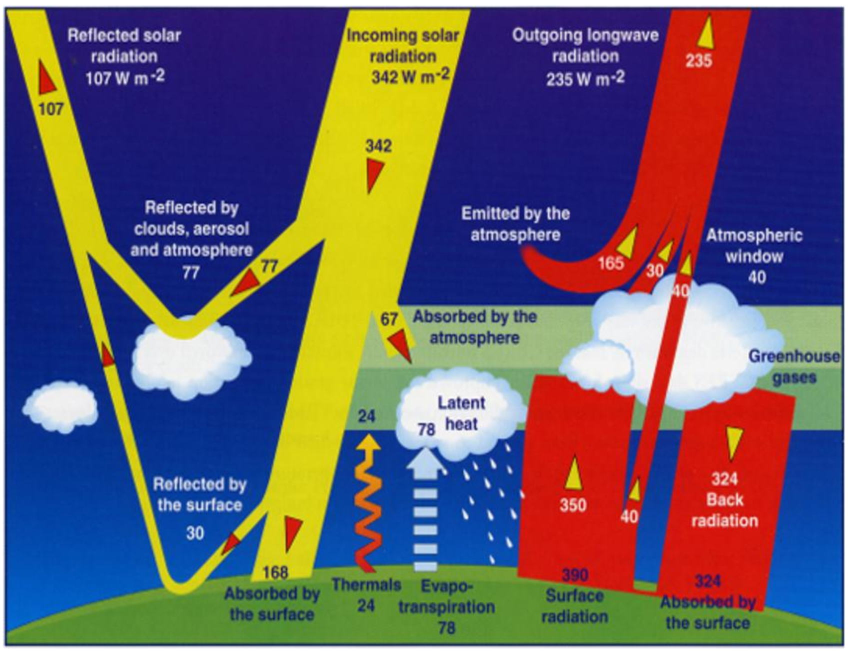



***Figure 1.*** *Energy balance diagram showing the flow of energy (in Watts per square meter, or W/m$^2$) from the Sun, from Earth's surface, and through the atmosphere. Yellow regions show incoming solar (shortwave) radiation. Red regions show outgoing infrared (longwave) radiation.*

**1A. (i) What three values illustrate the balance between incoming and outgoing energy?** Set up an equation with these three values illustrating this balance.

*Incoming solar radiation (342 W/m$^2$) - reflected solar radiation (107 W/m$^2$) = outgoing longwave radiation (235 W/m$^2$).*

**1A. (ii) Where does the incoming solar radiation go?** Name the possible outcomes for incoming solar radation based on the diagram.

*Incoming solar radiation is either: reflected by clouds, aerosol and atmosphere; reflected by the surface; absorbed by the surface; or absorbed by the atmosphere*

**1A. (iii) What values illustrate the balance between energy absorbed at the surface and lost at the surface?** Select all that apply and demonstrate the two energy flows are equal with a calculation.

Energy absorbed at the surface:

a. Solar radiation: 168 $W/m^2$
b. Thermals: 24 $W/m^2$
c. Surface radiation: 390 $W/m^2$
d. Back radiation: 324 $W/m^2$
e. Evapotranspiration: 78 $W/m^2$

Energy lost at the surface:

a. Solar radiation: 168 $W/m^2$
b. Thermals: 24 $W/m^2$
c. Surface radiation: 390 $W/m^2$
d. Back radiation: 324 $W/m^2$
e. Evapotranspiration: 78 $W/m^2$

*Energy absorbed: (a) solar (168) + (d) back radiation (324) = 492 $W/m^2$*
*Energy lost: (c) surface radiation (390) + (b) thermals (24) + (e) evapotranspiration (78) = 492 $W/m^2$*

**1A. (iv)** The outgoing longwave radiation is made up of energy emitted by the atmosphere (165 $W/m^2$) plus energy emitted by the clouds (30 $W/m^2$), plus energy escaping through the atmospheric window (40 $W/m^2$).

**What components together account for the energy emitted by the atmosphere plus that emitted by the clouds?** Select all that apply. *Hint: think about which components need to be added OR subtracted.*

a. Energy absorbed by the atmosphere coming in: 67 $W/m^2$
b. Thermal energy to the atmosphere: 24 $W/m^2$
c. Evapotranspiration to make clouds: 78 $W/m^2$
d. Energy reflected by the surface: 30 $W/m^2$
e. Surface radiation (not including atmospheric window): 350 $W/m^2$
f. Surface radiation (including atmospheric window): 390 $W/m^2$
g. Back radiation: -324 $W/m^2$

*a, b, c, e and g:*
*67 (absorbed by atmosphere coming in) + 24 (thermal energy to the atmosphere) + 78 (evapotranspiration to make clouds) + (350 – 324) (the net long-wave radiation)*

**1B. (i) How would an increase in albedo, defined as the amount of shortwave energy reflected by the Earth (surface + atmosphere), affect the energy balance in Figure 1?** Write 'increase', 'decrease', or 'not change' in the blank.

a. Shortwave energy absorbed by Earth's surface (168 $W/m^2$) would decrease.
b. Longwave energy input to the atmosphere by surface radiation (350 $W/m^2$) would decrease.
c. Incoming solar radiation (342 $W/m^2$) would not change.
d. Earth's average surface temperature would decrease.

**1B. (ii)** Table 1 (below) gives the value of albedo for various surfaces. **Which of the following would cause a decrease in Earth's surface albedo?** Select all that apply.

| Substance or Surface | Albedo |
|---|---|
| Whole Earth average | 0.3 |
| Fresh snow | 0.8-0.9 |
| White paint | 0.5-0.9 |
| Sea ice | 0.5-0.7 |
| Desert sand | 0.4 |
| Green grass | 0.25 |
| Bare soil | 0.17 |
| Dark roofs | 0.1-0.2 |
| Conifer forest | 0.08-0.15 |
| Open ocean | 0.06 |
| Fresh asphalt | 0.04 |

***Table 1.*** *Albedo of various substances and surfaces, defined as the fraction of solar radiation reflected. (Data from Akbari et al., 2012)*

a. Allowing the size of the Sahara Dessert to increase and take over forests
b. Melting sea ice
c. Replacing a grassy field or forested park with a parking lot
d. Planting grass in open fields of bare soil
e. Sea level rise increasing the surface area of Earth's oceans
f. Painting roofs of buildings white

*Choices (b), (c), and (e) are correct. Choices (a), (d), and (f) would cause an increase in albedo*

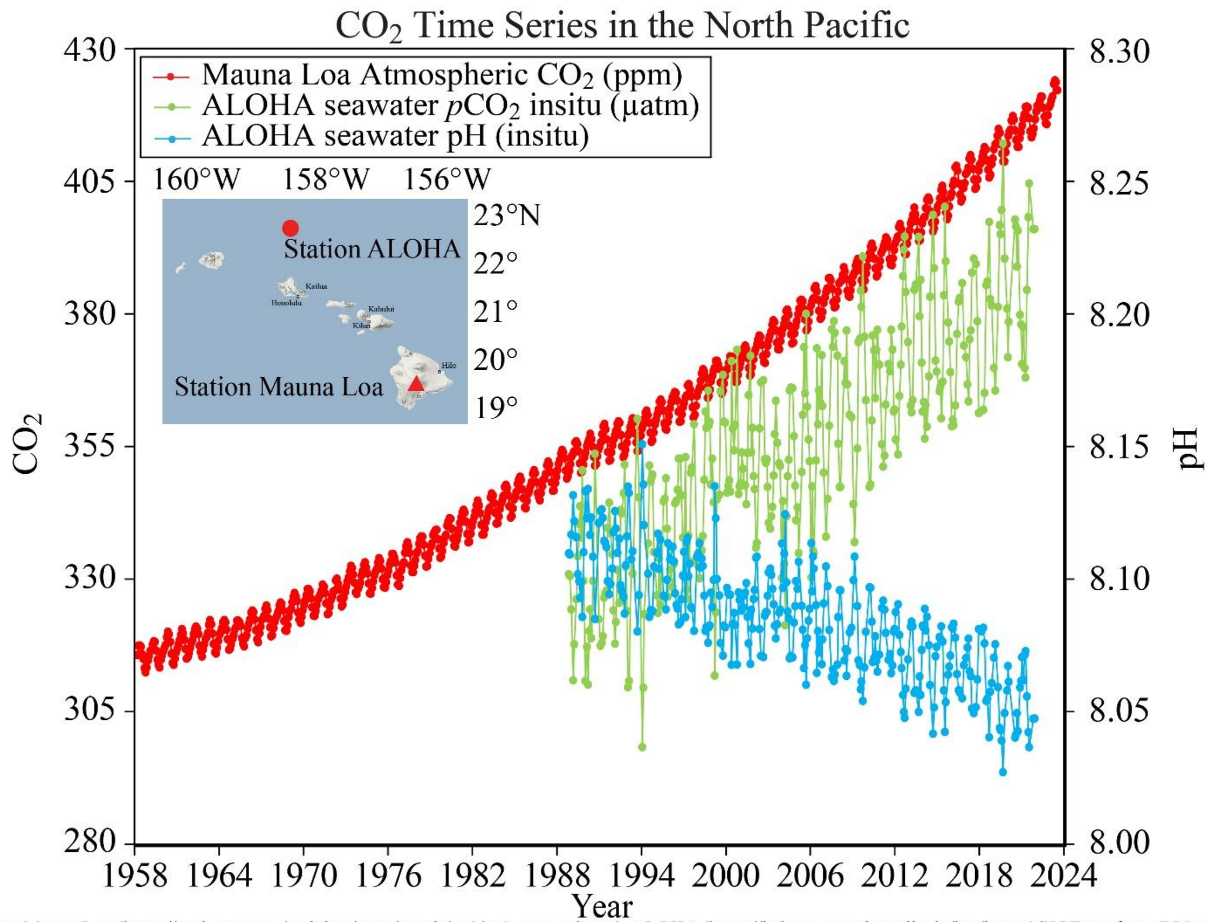
**1C. TRUE OR FALSE (correct the statement if false):** Greenhouse gases largely absorb incoming solar (shortwave) radiation.

*FALSE. Greenhouse gases largely absorb outgoing longwave (infrared) radiation.*
*OR FALSE. Greenhouse gases absorb a small amount of incoming solar (shortwave) radiation.*

*Extra note: The only greenhouse gas that does significantly absorb in the shortwave (solar) part of the electromagnetic spectrum is ozone, which is why it is such an important component of our atmosphere that protects against excessive ultraviolet solar radiation.*

## 2. Recent Climate Change and Impacts

Figure 2 (next page) shows measurements of atmospheric $CO_2$ from the Mauna Loa Observatory station in Hawaii starting in 1958. Seawater pH and seawater partial pressure $CO_2$ have also been measured at the ALOHA station off the coast of the Hawaiian island O'ahu (see inset map) since approximately 1990, as shown in Figure 2.




Data: Mauna Loa (https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_mm_mlo.txt) ALOHA (https://hahana.soest.hawaii.edu/hot/hotco2/HOT_surface_CO2.txt)
ALOHA pH & $pCO_2$ are calculated at in-situ temperature from DIC & TA (measured from samples collected on Hawaii Ocean Times-series (HOT) cruises) using co2sys (Pelletier, v25b06) with constants: Lueker et al. 2000, KSO4: Dickson, Total boron: Lee et al. 2010, & KF: seacarb

***Figure 2.*** *Measurements of atmospheric $CO_2$ concentration (red line) from the Mauna Loa Observatory station from 1958 to 2024, seawater partial pressure of $CO_2$ in µatm (green line) and seawater pH (blue line) from station ALOHA from 1990 to February 2024. Source: National Oceanic and Atmospheric Administration Pacific Marine Environmental Laboratory Carbon Program.*

**2A.** The record of atmospheric $CO_2$ concentrations from Mauna Loa, Hawaii shows both a long-term trend from 1958-2024, as well as short-term (intra-annual) variability. **Explain what causes intra-annual variability in atmospheric $CO_2$ in 2-3 sentences.**

*Intra-annual variation in the record of atmospheric $CO_2$ corresponds to changes in the seasonal uptake of $CO_2$ by plants in the Northern Hemisphere. During spring/summer, plants take up $CO_2$ during photosynthesis and atmospheric $CO_2$ levels drop, reaching a minimum in early Fall. Conversely, during fall/winter, more $CO_2$ is being released to the atmosphere from plant respiration and the decomposition of leaf litter/dead plant material, and atmospheric $CO_2$ concentrations reach a seasonal maximum in early spring.*

**2B. (i) Using Figure 2, estimate the annual rate of change for atmospheric $CO_2$ concentration (in ppm $CO_2$ per year) at the Mauna Loa Observatory station for the time period from 1958-1976.** Show your work, then select the multiple choice answer that is closest to the value you calculated.

a. 0.05 ppm/year
b. 0.1 ppm/year
c. 0.8 ppm/year
d. 1.6 ppm/year
e. 2.3 ppm/year

*(c) is correct*
*$CO_2$ in 1958 = roughly 315 ppm; $CO_2$ in 1976 = roughly 330 ppm*
*330 ppm – 315 ppm / 18 years = 0.8 ppm/year.*

**2B. (ii) Now estimate the average annual rate of change for the time period from 2006 to 2024 (Column A). How many times larger is this annual rate of change in atmospheric $CO_2$ compared to the period from 1958-1976, roughly half a century earlier (Column B)?** Show your work, then select the multiple choice answers that are closest to the values you calculated. *Note: use the multiple choice value selected in Column A to determine the rate for Column B.*

| A | B |
|---|---|
| a. 0.05 ppm/year | f. It is the same rate |
| b. 0.1 ppm/year | g. The 2006-2024 rate is 1.5 times larger |
| c. 0.8 ppm/year | h. The 2006-2024 rate is 3 times larger |
| d. 1.6 ppm/year | i. The 2006-2024 rate is 4 times larger |
| e. 2.3 ppm/year | j. The 2006-2024 rate is 5 times larger |

*(e) and (h) are correct*
*$CO_2$ in 2006 = roughly 380 ppm; $CO_2$ in 2024= roughly 422 ppm*
*422 ppm – 380 ppm / 18 years =2.33 ppm/year.*
*2.33 / 0.8 = 2.9. The annual rate of CO2 increase is ~3 times higher than it was in 1958-1976!*

**2C.** As the concentration of $CO_2$ in the atmosphere increases, the concentration of $CO_2$ dissolved in seawater also increases, causing a change in pH. **Based on Figure 2, what can we conclude about the relationship between atmospheric $CO_2$ concentration and seawater acidity?** Note that a decrease in pH means the water is becoming more acidic while an increase in pH means the water is becoming more basic.

As atmospheric $CO_2$ increases, seawater becomes more acidic (pH goes down)

**2D.** The ocean is a major carbon "sink" and currently absorbs about 31% of the $CO_2$ released by anthropogenic sources into the atmosphere (Gruber et al., 2019)[1]. This means the rate you calculated in question 2B is actually only 69% of what it would be if the ocean were not able to absorb atmospheric $CO_2$. Unfortunately, the ocean's ability to absorb $CO_2$ is finite, and has been declining over the past decade as warmer oceans have less capacity for $CO_2$ uptake. **Based on**

**your *multiple choice answer* to 2B (ii), what would the concentration of atmospheric $CO_2$ be this year (2025) if ocean absorption of atmospheric $CO_2$ had all of a sudden gone from 31% per year to 0% per year starting in 2006 (when atmospheric $CO_2$ was 380 ppm)?** Compare this value to the observed atmospheric $CO_2$ that we have today of 425 ppm (Oct 2025 average).

*Without the ocean being able to act as a sink, the rate of $CO_2$ increase in the atmosphere per year will increase to: 2.3/0.69 = 3.33 ppm per year starting in 2006.*
*The concentration of atmospheric $CO_2$ in 2006 = roughly 380 ppm so we have:*
*380 ppm + 3.33 ppm per year * 19 years = about 443 ppm.*
*The atmospheric concentration of $CO_2$ would be 18 ppm higher today (443-425).*

**2E.** Figure 3 shows the flux of carbon (in petagrams of C per year) going into the atmosphere measured at the Mauna Loa Observatory in Hawaii (blue line), in ice core bubbles from Antarctica (black line) and released by the burning of fossil fuels (red line).

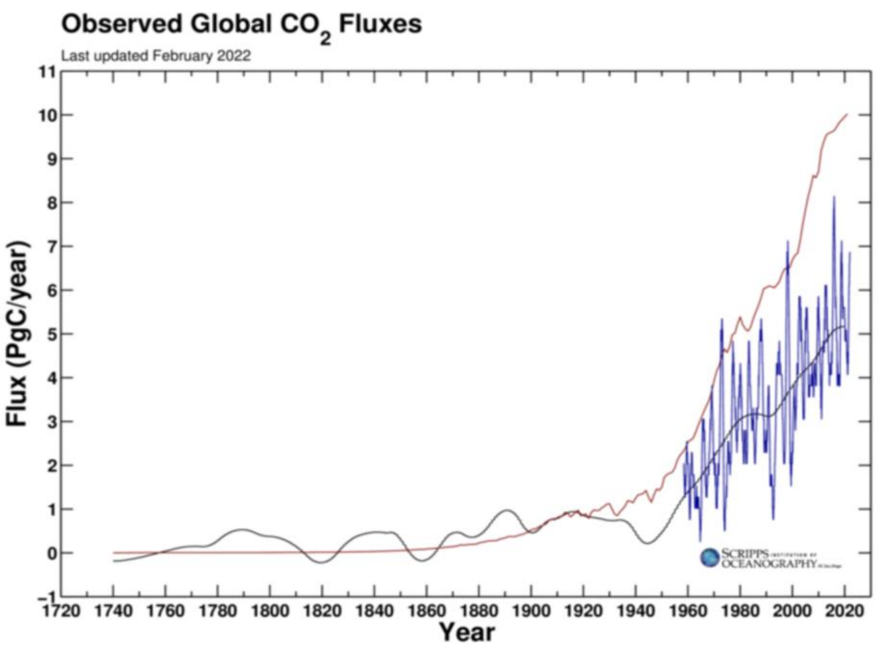



***Figure 3.*** *The annual flux of $CO_2$ (mass per year) is calculated using the yearly increase in atmospheric $CO_2$ concentration. Data from Scripps Institute of Oceanography.*

**2E. (i) Based on Figure 3 (above), estimate the difference between the flux of carbon released by fossil fuels and the flux of carbon measured in Antarctica in 2020.** Give your answer in kilograms of carbon per day and write your answer in scientific notation (1 petagram = $1 \times 10^{15}$ grams).

*Flux measured in Antarctica = 5.2 Pg C/year*
*Flux released by fossil fuels = 10 Pg C/year*
*Total approximate difference = 10 - 5.2 = 4.8 Pg C/year*

*4.8 Pg C/year * $1 \times 10^{12}$ kg/Pg C = $4.8 \times 10^{12}$ kg C/year x 1 yr/365 d = $1.32 \times 10^{10}$ kg C/day*

**2E. (ii)** We already learned that some of the carbon released by the burning of fossil fuels is being absorbed by the ocean. Where else could it be going? **Provide another reasonable carbon sink on Earth where the excess carbon you calculated in 2E. (i) is going.**

*Another sink is the biosphere, where plants will take in carbon dioxide for use in photosynthesis. This is why deforestation and other land cover changes contribute to an increase in global temperatures.*

**2F.** In Prof. Helfand's lecture, you learned about the use of isotope ratios to discern the source of atmospheric carbon. Figure 4 below shows the ratio of $^{13}C/^{12}C$ in atmospheric $CO_2$ (relative to a standard), which is denoted as $\delta^{13}C$. The black line shows the value recorded at the Mauna Loa Observatory, Hawaii, and the red line shows the value recorded at the South Pole, Antarctica.

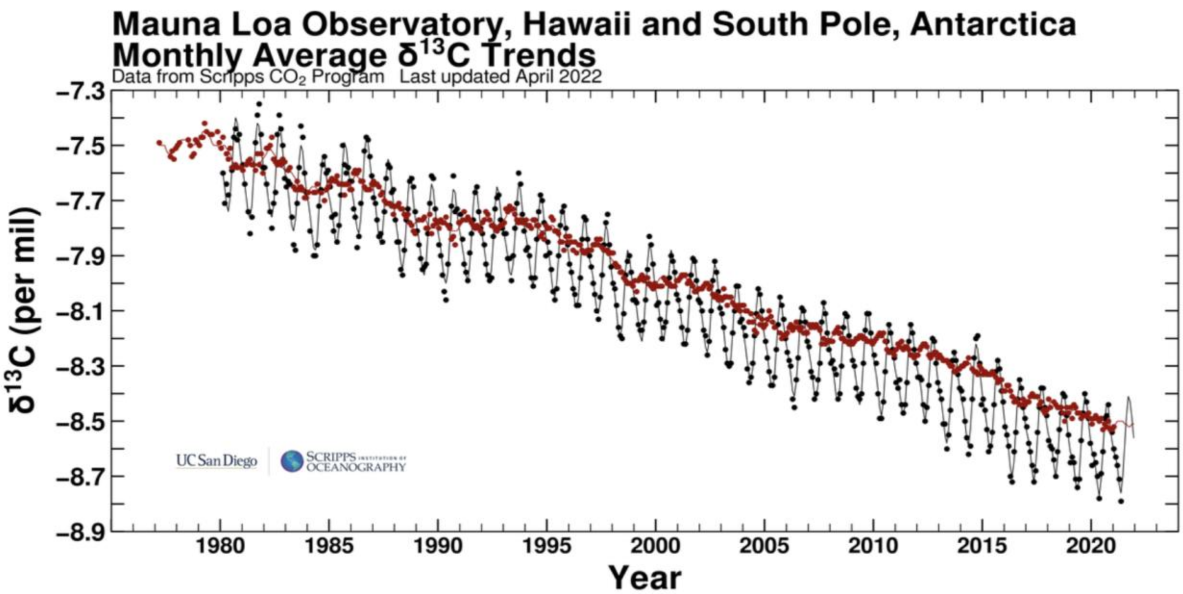



***Figure 4.*** *Average monthly $\delta^{13}C$ records of atmospheric $CO_2$ from Mauna Loa Observatory, Hawaii (black dots/curve) and South Pole, Antarctica (red dots/curve) from 1975 to 2022. (Note: $\delta^{13}C$ represents the ratio of $^{13}C/^{12}C$ relative to a standard and is recorded in parts per thousand or "per mil"). Source: Scripps $CO_2$ program.*

**2F. (i)** Changes in $\delta^{13}C$ represent the ratio of $^{13}C/^{12}C$ relative to a standard, so a higher (less negative) $\delta^{13}C$ corresponds to a higher $^{13}C/^{12}C$ ratio and a lower (more negative) $\delta^{13}C$ corresponds to a lower $^{13}C/^{12}C$ ratio. **Based on what you learned in lecture, why does a decline in $\delta^{13}C$ through time provide support that fossil fuels are the source of increasing atmospheric $CO_2$?** Explain in 2-3 sentences.

*During photosynthesis, plants preferentially take up and store the lighter $^{12}C$ isotope of carbon. Fossil fuels are from old plants and thus have a low ratio of $^{13}C/^{12}C$ (a negative $\delta^{13}C$, somewhere around -25 per mil). Therefore, adding carbon from fossil fuels to the atmosphere would reduce the $\delta^{13}C$ signature of the atmosphere, which is consistent with the observed trend in $\delta^{13}C$.*

**2F. (ii) Which location shows greater intra-annual variability in $\delta^{13}C$ value of atmospheric carbon dioxide: the South Pole, Antarctica or Mauna Loa, Hawaii?** Circle the correct answer and in 2-3 sentences, speculate as to what may be responsible for this difference.

*The $\delta^{13}C$ values of atmospheric carbon dioxide measured at the Mauna Loa, Hawaii site shows more intra-annual variability than the values for carbon dioxide measured at the South Pole. This is because the intake of carbon dioxide by plants in the spring and summer vs. release in fall and winter is more dramatic in the Northern Hemisphere, due to larger land masses and prevalence of seasonal vegetation. Plants preferentially take in the $^{12}C$ isotope of carbon more in the spring and summer, increasing the $\delta^{13}C$ value of the carbon dioxide left in the atmosphere at this time.*

[1] *Nicolas Gruber et al. The oceanic sink for anthropogenic $CO_2$ from 1994 to 2007. Science* ***363****,1193-1199 (2019). DOI:*10.1126/science.aau5153